

Technology News

Robot that eats and poos

Researchers at Bristol Robotics Lab in the UK have created an eating robot, powered by 48 small microbial fuel cells. EcoBot III, also known as BREADbot (BREAD is short for 'bio-regulation and energy-autonomy with digestion'), has an artificial gut that allows it to survive up to seven days, feeding and 'watering' itself without human assistance.

The robot, when guided towards its food by a light source, feeds on a nutrient-rich diet from the sewage. The bacteria process the sewage mash and the microbial fuel cells recycle the food, producing hydrogen and electricity in the process. To maintain power generation, the robot needs regular drinks of water, which it gets from a special waste chamber.

EcoBot III does not require human assistance to clean up after meals. Once every 24 hours, the recycled food is

gathered via gravity feed in a litter tray, making EcoBot III the first poop-bot of its kind.

Thought-controlled in-flight entertainment

The global aerospace industry was treated to a sneak peek into the future of in-flight entertainment as

car. But, with researchers working on a technology called 'non-visual interfaces,' this far-flung notion could soon be a reality. The National Federation of the Blind, USA, and Virginia Tech are likely to present the prototype of such a car in 2011.

The car will use sensors to guide the visually challenged driver through



InteraXon thought-controlled in-flight entertainment system

Toronto's InteraXon demonstrated a prototype of their thought-controlled in-flight entertainment system to overwhelmingly positive responses. The system was shown at *On the Wings of Innovation*, a global aerospace symposium organised by the Ontario Aerospace Council and the Ontario government. The demonstration included brainwave-controlled games and applications designed for use at heights of 10,668 metres (35,000 feet).

The system measures the brain's electrical output and sorts it into waves. It reacts to alpha waves, associated with relaxation, and beta waves, associated with concentration. As the users relax or focus their thoughts, their brainwaves become the interface with which users control the game.

Car for the blind

It's hard to imagine a visually impaired person behind the wheel of a

traffic by transmitting information about his nearby vehicles or objects. Many other innovations will make it easy for him to operate the car. Vibrating gloves, for instance, will help in communicating the necessary information needed to avoid collisions.

Another interface, called AirPix, is a tablet about half the size of a sheet of paper with multiple air holes. Compressed air coming out of it will help inform the driver of his surroundings, essentially creating a map of the objects around the vehicle. It would show whether there was another vehicle in a nearby lane or an obstruction in the road.

The model to be demonstrated next year will be a modified Ford Escape sports utility vehicle.

Car that tweets

AJ (short for American Journey 2.0), a future model of Ford Fiesta, can tweet. It is a test ground for the company



EcoBot-III robot